14. Natural Language Processing (NLP)

Is a branch of artificial Intelligence that helps computers read, write, and speak human languages.

-It combines computer science, linguistics and machine learning. It turns raw text or speech into data a machine can process.

-It breaks down grammar, context and meaning
-It allows systems to reply with human-like text or speech.

Three real-world applications
1. Spam Filters: Email programs scan incoming messages using NLP to spot tricky words, patterns, and intents, moving junk email out of your main inbox.

2. Search and Social Media

3. Voice Assistants

15. Large Language Model (LLM)

Is an advanced type of artificial Intelligence trained on massive amounts of text data to understand, summarize, and generate human-like language.

-It uses deep learning and a special neutral network design called a transfomer.

-It works by looking at words or parts of words called tokens and predicting the next most likely word in a sentence.

-It contains billions of adjustable settings called parameters that help it learn deep context and grammar.

How is it different from Standard Machine Learning Model?

1. Scope and data: ML often handles structured data (like numbers in a spreadsheet) to do specific tasks like predict houses or sort emails into spam

-AN LLM is a specialized subset of deep learning focused strictly on unstructured human language and text patterns at a massive scale

2. Size and Complexibility
Standart ~~LLM~~ ML models use thousands or millions of parameters, while LLMs use billions or trillions of parameters.

3. Flexibility
- Standard ML models usually need manual feature engineering and are built for one exact task.
- An LLM can adapt to many different text-based tasks (like writting, coding or translating) without being retrained from Scratch.

Two well-known LLMs

1. Open AI GPT - A powerful family of generative models (powering tools like ChatGPT)

- Used for writting emails, essays, code.
- Answering questions and explaining things

2. Google Gemini - Is a multi-modal AI model built to process text, images, audio, video and code all at the same time.

- Used for analyzing documents, powering Conversational assistants, summarizing research, building interactive software tools.

16. Generative AI

- Is AI that can create original content such as texts, image, video, audio or Software code in response to a user's prompt or request.

How it works at a High level

- Training on vast data - models are trained on massive datasets containing books, articles, images, or code to learn underlying patterns, rules, and structures.

- Pattern recognition - AI breaks inputs down into smaller pieces called tokens
- Prediction and generation - when you give it a prompt, the model does not search for exact pre-made answer; instead, it predicts and generates new, contextually relevant output piece by piece based on the patterns it learned.

3 tools that use Gen AI

1. ChatGPT - A conversational text assistant that answers questions, write essays, summarize documents, and helps with brainstorming or coding

2. DALL-E 3 - An image-generation model that creates unique, high-resolution illustrations, artwork, and photos from descriptive text prompts.

3. GitHub Copilot - An AI pair programmer that assists software developers by auto-completing code, suggesting entire functions, and helping debug errors.

17. Training Dataset

Is a collection of information - such as text, images, or audio - used to teach an AI model how to recognize patterns, make decisions or perform specific tasks.

The quality of training is important because;

- Direct foundation: AI models learn directly from the examples they are given, their outputs reflect the exact signals, patterns and flaws present in that output.

- Generalization: High-quality data helps the model understand underlying concepts so it can perform accurately to new, real-world situations it has never seen before.

- Reliability and Trust: clean, well-labelled, and diverse data reduces random erros and ensures the system behaves consistenly under normal conditions.

If the data is incorrect it could result in;
- Error Propagation: Mislabeled or factually incorrect examples teach the model false rules, leading to confidente but completely wrong predictions at scale.

- There could be Bias Amplification: If the training data favors a specific group or viewpoint, the AI absorbs that prejudice and turns it into systems.

- Blind spots and Failures: A lack of diversity cases means the AI will break down, fail catastrophically when faced with unexpected real-world inputs.

- Project Abandonment: Poor data quality can cause entire AI initiatives to produce useless results or regulatory penalties, leading organizations to abandon the project.

18. Computer Vision

Is a field of artificial intelligence that trains machines to extract, interprete, and understand meaningful information from digital images and videos.

I. Machine see and interprete images;

- Pixel Breakdown: A computer sees an image as a massive grid of numbers representing pixel values (colour and brightness).

- Feature Extraction: Algorithms scan the grid to find basic edges, shapes, and colour gradients

-Deep learning (CNNs): specialized neutral networks (like convolutional Neutral Networks) process these features through layers to recognize complex patterns.

-Classification and Context: The system matches detected patterns against trained data to label objects, locate boundaries, and understand the full scene context.

Real-world use cases:

1. Healthcare: X-ray/MRI analysis - they detect cancer, fractures faster than doctors.

2. Agriculture: Drones scan crops for disease, count fruits, spray only when needed.

Section C: APIs, Tools & Developer Concepts

1. API (Application Programming Interface) is a computing interface which defines interactions between multiple software intermediaries. It defines the kinds of calls or requests that can be made, how to make them, the data formats that should be used, the conventions to follow.

Using a banking app to explain API:
The banking app does not hold money - the bank's server does.
- When one taps 'Check Balance', the app asks the API. The API goes to ~~check~~ the server, gets the balance, and brings it back to show you.

An API is ~~the~~ secure messenger between your app and the bank.

Why are APIs important in the modern world?

- Seamless Communication: They allow competely different technologies and programming languages to talk to each other without friction.

- Reusability and speed: Developers do not have to reinvent the wheel, they can plug into existing services (like using Google Maps inside a food delivery app).

- Automation: They let software systems trigger actions across multiple platforms automatically, reducing manual work.

- Scalability: Companies can update or expand their backend systems or ~~adding~~ add cloud services without breaking the user-facing apps.

- Security: They act as a controlled gatekeeper, exposing only the necessary data while keeping internal systems details hidden and secure

20. An API key

- Is a unique alphanumeric string used to identify and authenticate a Software application or developer when making a request to an API

Purpose of an API key

- Authentication: Verifies which application or Project is making the request.

- Access Control: Blocks anonymous or unauthorized traffic from using a service.

- Rate Limiting: Controls call volumes to prevent systems overload:

- Usage Monitoring: Tracks traffic patterns, logs activity, monitors resource consumption.

How to store and Protect an API key

- Rotate keys regularly
- Use environment variables
- Leverage Secrete managers
- Restrict Key Permissions
- Implement IP Restrictions

-If the keys are exposed publicly; the would be Unauthorized data acces, financial loss, system compromise and reputational damage

24. Rest API

Is an architectural style for building networked applications, allowing software systems to communicate with each other over the internet.
It relies on a client-server model where data is treated as "resources" (like a user, product, or image) that can be accessed and manipulated using standard web protocols.

REST stands for Representational State Transfer

In a REST API, communication is stateless, and operations correspond to the HTTP methods that map directly to basic CRUD (Create, Read, Update, Delete) actions

The Main HTTP Methods
-GET (Read): Retrieve data from the serve. Making a GET request fetches a specific resource or a list of resources without changing anything on the Server.

-~~Post~~ POST (Create): Sends new data to the server to create a resource. for example, Submitting a form to Sign up a new ~~the~~ user

or adding a new transaction triggers a POST request.

- PUT (Update/Replace): Updates or completely replaces an existing resource on the server. If the resource does not exist, it can sometimes be used to create one at the specific location.

- DELETE (Delete): Removes a specified resource from the server. Sending a DELETE request permanently discard the targeted data.

22. JSON

- JavaScript Object Notation) is a lightweight, text-based data format used for storing and exchanging data.

Why is it Commonly used for APIs

- Lightweight and Compact: JSON uses a highly minimalist syntax. JSON files have smaller file sizes, resulting in faster data transmission over networks.

- Human-Readable: It structures data using intuitive key-value pairs (for objects) and ordered lists (for arrays). This makes it straightforward for developers to debug and inspect API requests.

-Easy to Parse: Almost all modern programming languages have built-in libraries or native methods to easily convert (parse) JSON strings into usable native data structures (like maps, dictionaries, or objects) with minimal code

-Universal Compatibility: Any platform can send or receive JSON because it is strictly text

-~~Universal~~

23. Prompt Engineering

Is the practice of designing, structuring, and refining the text inputs (prompts) you give to an artificial intelligence tool to get the most accurate, useful, and relevant response.

Why it is an important skill

-It reduces guesswork: clear instructions stop the AI from making random assumptions or guessing your goal

-Saves time: A well-crafted prompt gets the right on the first try instead of needing five rounds of corrections

-Improves Control: You can dictate the exact tone, format, length, and perspective you need.

-Acts like delegation: Treating the prompt like a professional brief helps the AI function as as a capable assistant rather than basic search engine

Two examples

1. Poor Prompt

"Tell me about banking"

·Its too vague, the AI does not know if you want history, types, how it works, or for what level.

Improved version

"Explain how banking mobile app works for a first-year University student. Use simple language, no code, and give one real-life example of checking a balance"

2- Poor prompt

"Fix my data"

There is no context,

Improved version

"I have a Google Sheet called 'Shop_Performance_2019_2020 with columns Payment_status and Product_name. Write a COUNTIF formula to count how many orders are "Paid" and explain how it works in one sentence"

24. MAP Function

Is a built-in higher-order function used in programming and data processing to apply a specific transformation function to every item in a collection (such as a result or array) without using an explicit loop.

What it does in Programming & data Processing

- It transforms data: It takes an input and an instruction (a function), then creates a matching output collection where every single element has been modified by that instruction

- Maintains structure: The resulting collection always has the same number of items as the the original collection.

- Promotes immutability: It generates a brand-new dataset with the changes applied, it does not change or overwrite the original data.

- Enables Parallel Processing: In large-scale data systems, mapping is the first critical step where massive datasets are split up so thousands of computers can transform pieces of the data simultaneou

Python example:

How to use maps() to square a list of numbers

```
#1. Define the transformation rule

def square(number):
    return number**2

#2. Create the initial collection
numbers = [1, 2, 3, 4, 5]

#3. Apply the map function

#(Note: In Python, map returns an iterator, so we convert it to a list to print it)

squared_numbers = list(map(square, numbers))

print(squared_numbers)
# Output: [1, 4, 9, 16, 25]
```

25. Cloud Computing

Is the delivery of on-demand computing services - including servers, storage, databases, networking, and software - over Internet on pay-as-you-go basis.

You rent access to infrastructure hosted by cloud provider.

Three main cloud Services

Iaas - Intrastructure as a Services

1. Iaas provides basic, virtualized computing resources over the internet, including ser storage, networking. You rent the raw hardware and manage your own operat systems, databases, and application.

Example : Amazon EC2 (Elastic Compute cloud)

2. Platform as a Service (PaaS)

PaaS delivers a ready-to-use cloud environment designed for developers. The provider manages the underlying servers, storage, and operating systems, allowing you to focus entirely on writi testing, and ~~develo~~ deploying your code.

3. Sofware as a Services (SaaS)

SaaS delivers fully-functional, ready-to- software applications over the web on a subscription basis. The provider handles all maintenance, updates, security, and infrast you just log in and use the application

Example : Google workspace

26. Version Control System (VCS)

Is a tool used used in software development and collaborative projects to track and manage changes to use source code.

Git is a free, open-source distributed version control system designed to handle everything from small to large projects with speed and data integrity. It is "distributed" because every user has a complete local copy of the Project and its entire change history.

Why Git is used widely

- For collaboration: Multiple people can work on the same project at the same time without overwriting each other's work.

- History Tracking: It records who made changes, when they made them, and why, making it easy to see project evolution or find bugs.

- Safely: If a file breaks, you can easily roll back or revert to an older, working version.

- Reproducibility for Data Analysts: Analysts use Git to track changes in data scripts, notebooks, and models, ensuring their analysis can be reliably reproduced.

Commit is a saved snapshot of your project files at a specific point in time. It records changes to your codebase, who made the changes, when it happened, and a unique ID(hash). Purpose is to track project history and return to previous state if something breaks

Branch

- Is a movable pointer or reference to one of the commits. It represents a separate stream of development in your project. Purpose is to allow you to build new features or fix bugs safely without changing the stable main code.

27. Jupyter Notebook

Is an open source web application that allows data scientists to create and share documents that include live code, equations, and other multimedia resources.

The programming language it is most commonly used with is Python.

Why is it popular among Data Analysts & Scientist

1. Cell-by-cell Execution – can run small block (cells) of code independently.

2. Inline Visualizations: It immediately shows or generate charts, tables, and graphs below the code, making trends instantly visible

3. Computational Storytelling: It blends analytical code with a narrative text (using markdown) and math formatting (via LaTeX).

3. Seamless Integration with Data Libraries
It accommodates standard Python data science libraries like Pandas for data manipulation, Numpy. for numerical computing, Matplotlib/Seaborn for data plotting, and Scikit-learn for machine learning prototyping

4. Easy Sharing and Collaboration: Notebooks Can be easily shared as raw files or converted into clean, production-free formats like HTML, PDF to share with non-technical team members.

28. Python

-Is a Programming language that is widely used in web applications, software development, data science, and machine learning (ML)

Why does it dominate

Because of its simplicity and readability
- it emphasizes code readability by using English words instead of complex symbols, making it highly accessible.

- It has Vast Ecosystem of Libraries
- Seamless workflow intergration
- Massive Community Support

Four common Python Libraries

1. Pandas - Used for data manipulation and analysis.

2. Numpy : Used for numerical and scientific computing

3. Matplotlib : Used for data visualization

4. Scikit-learn : For traditional machine learning.

SECTION D: Data Ethics, Governance & Careers

29. Data Privacy:

Is fundamental to ensuring privacy and security in a digital environment.

Why is it important?

- To prevent Identity Theft: Stops criminals from using exposed records for fraud.
- Protects Personal Freedom: Blocks intrusive tracking and mass surveillance.
- Builds Trust: Helps customers feel safe interacting with businesses.
- Ensures Legal Compliance: Avoids heavy fines under laws like GDPR

General Data Protection Regulation (GDPR)

- It is a comprehensive European Union (EU) law on data protection and privacy that went to effect on May, 25, 2018.
It regulates how the organizations collect, use, store and share data of individuals.
- It applies directly across all 30 EU and EEA member countries, but it also applies globally to any organization that handles the data of individuals located in these regions

South Africa has POPIA Act (Protection of Personal Information Act)

30. Data Ethics

Is the branch of ethics that studies and evaluates moral problems related to data generation, collection, curation, processing, sharing, and use, alongside artificial intelligence and algorithms, to promote responsible and fair digital practices.

Three examples of unethical use of data/AI

1. Unauthorized Biometric Scraping (Clearview AI)

A tech company scraped billions of facial images from public social profiles without user consent. ~~to build~~

It violates individual privacy

2. Algorithmic Discrimination in hiring (Biased AI)
- An automated tool (hiring tool) was trained on historical company resume from a male-dominated industry, leading the AI to penalize resumes that contain words like "women".

- It weaponizes historical bias, creates inequality
- Gender inequality.

3. Mass behavioral harvesting (e.g Cambridge Analytica)
- Harvesting of personal data from millions of social media users without explicit consent through quiz app and repurposed to micro-target political advertisements and manipulate voter behavior.

- It breaches informed consent and misuses personal information for covert psychological manipulation.

~~33~~ 31. Data Governance

Is a governance framework that defines how organizations manage data assets across their lifecycle. Its purpose is to improve data integrity, ensure compliance with regulations.

1. Policies and standards: Are documented rules that dictate how data is handled, ~~stored~~ shared, retained, and structured across the organization.

2. Data ownership and Stewardship: Clear assignments of accountability.
Data owners manage specific business domains, while data stewards ensure day-to-day quality and compliance.

3. Data quality and Integrity
- Setting standards for accuracy, completeness, consistency, as well as monitoring for duplication and errors

4. Security and Compliance Controls
- A data governance must define how access is granted, how long records are retained, and how activity is logged for audit purposes. This is to reduce the risk of unauthorized access, data leaks, or policy violations.

5. ~~Data~~ Architecture and Technology
Defines how an organization collects, stores, transforms, manages, and uses its data assets.

6. Compliance and Auditing
- Measures to meet legal and regulatory requirements (such as GDPR or HIPAA) and continuous monitoring to evaluate and improve ~~governmen~~ goverrance maturity.

7. Data Lineage and Cataloging
Tracking where data comes from, how it changes, and where it moves across systems.

32. Bias AI
Refers to systematic discrimination embeded within AI systems that can reinforce existing biases, and amplify discrimination, prejudice, and stereotype.

What it means for an AI model to be Biased

-It means its outputs or decisions contain systematic, repeatable errors that unfairly disadvantage or priviledge specific group. This usually happens because of data used to train the AI, algorithm bias, human decision bias and generative AI bias.

Real-world example of AI bias

Amazon's experimental hiring algorithm

- The computer model was trained on ten years of historical resume submissions. Because the tech industry historically employed more men, the training data contained predominantly male applications.

- The algorithm learned to penelize resumes that included words associated with women's organizations and dowgraded female applicants while favoring male-associated language.

Potential Consequences

- Reinforcing Inequality - divides society
- Legal and business risks - damages companies reputation, and could cause legal penalties.
- Safety and health harms - could lead to dangerous misdiagnoses or wrongful arrests.

Role	Role	Skills	Tools
~~Data~~ Analyst ~~Data~~	Interprete past data, create dashboards & report trends to guide business choices	Data cleaning, basic statistics, data visualization, business communication	SQL, Exc Tableau, Po BI, basic Python
Data Scientist	Build machine learning models & run experiments, to predict future events	Advanced mathematics, statistics, predictive modeling, pattern recognition	SQL, Pyth Pandas, R Scikit-learn
Data Engineer	Design and maintain scalable data pipelines and storage systems	Data architecture, ETL, design, database management, cloud computing	SQL, Apac Airflow, Pyt Spark, Snow
AI Engineer	Design Intelligence applications using pre-trained foundation models and LLMs	Prompt engineering, RAG, API integration, system deployments	Python PyTorch, TensorFlow, Hugging Fac LangChain

34. Business Intelligence

Is a set of technologies, tools and practices that collect, organize, and analyze past and current data to help companies make informed decisions.

Popular BI Tools

1. PowerBI – A microsoft cloud service that connects easily with Office apps, Excel and various databases for interactive reporting. The visuals make it clear to a non-technical person to understand what is happening in the business and make informed decisions.

2. Tableau
A visual analytics platform know for processing large datasets quickly and building deep, customizable visual dashboards.

How the tools drive data-driven decisions

1. Automate data collection: They pull raw information from different sources – like sales, finance, and customer service – into one central place

2. Turn data into visuals – they change complex numbers into clear charts, graphs and live dashboards.

3. Spot Trends early

- They use data mining to find hidden patterns, recurring problems, or new mark opportunities.

4. Remove Guesswork
Leaders use real-time facts instead of intuiti to manage daily operations and long-term goals.

SECTION E: Personal Reflection

35. The concept that I found most interesting is NLP (Natural Language Processing). It was interesting to learn how computers is being helped by AI to read, write and speak human languages. I have always wondered how the likes of Siri work. And training dataset

36. The concept that I have found most difficult to understand is Map function. I had to watch YouTube videos (mastering Python's MAP Function) by Amir Clarkhi.

37. AI is not here to replace human beings, in order for one to empower themselv is to understand how AI works and you it to their advantage. If you do not know how

AI works, the outputs will not be accurate. AI still needs human beings to confirm the outputs.